

Higher Education Doesn't Always Mean Higher Potential



Ahsan Tebha

Dr. Young

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Contents

Abstract	2
Research Question	2
Methodology	2
Analysis.....	Error! Bookmark not defined.
Conclusion	4
References.....	4

Abstract

This study examines whether the type of institution a student attends or the major they choose has a larger impact on post-grad earnings. Using a dataset of 30,000 graduates nationally, I analyzed the added earnings 10 years after graduation over the base earnings of a high school graduate across 19 different majors. These majors were then categorized into 7 buckets. The type of College or University was categorized into 5 buckets: “For Profit”, “Regional”, “Mid-Tier”, “Top Public”, and “Ivy Elite”. I compared the variance of each factor using ordinary least squares (OLS) and analyzed the variance.

Major explained far more variation than university tier did. There was almost a 55% variance for major versus just a 6% variance on university tier in terms of potential earnings. Analyzing through a log scale, I found there was roughly a 77% impact of major compared to just about a 5% for university tier. The mix of grads was balanced across the tiers, so impact was mostly independent, and when factoring in cost, the impact did not change. There was a unique R Squared of .060 versus a .058 prior. When looking at the dollar amount, where the tier premium looked larger in the higher-paying majors, this did not materialize when using a log scale. This indicated a consistent percentage increase across all majors. For-profit universities scored the lowest in terms of value, and when using the for-profit as the base, all other tiers scored significantly higher, like regional schools being 39%. Mid-tier at 64%, top public getting 88%, and Ivy getting 139%. When looking through the lens of major with Biology being the baseline, the range was anywhere between -79% less(Arts) to +207%(electrical engineering).

These findings suggest that major has a more significant impact than type of school but this only accounts for about 60%, leaving unknown factors at about 40%. These unknowns could be family socioeconomic standing, networking in school, geographic location, student ability, and also how much a person earns as a high school graduate changes as well geographically, further skewing earning potential depending on the major.

Future work should consider earnings geographically by field and compare that to majors. Along with a more defined methodology for calculating high school earning potential, factoring in technological disruptions such as AI implementation and such, these results are more observational than causal.

Research Question

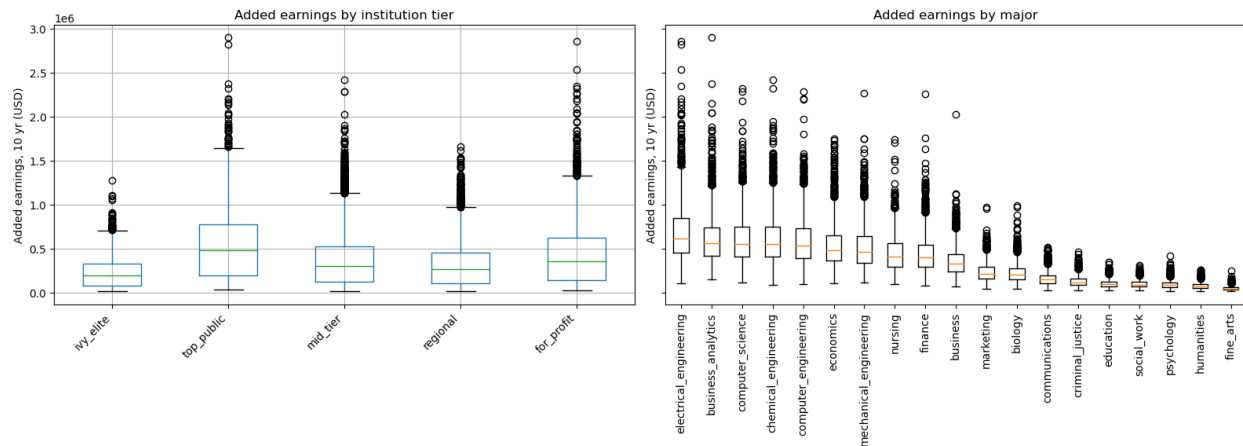
The research question I examined is, “Does the institution you attend have a larger effect on your earnings than the major you choose?”

Methodology & Analysis

The method I used was to find a dataset, which was from Kaggle¹. This dataset had 30,000 rows and had high school earnings as well as post-graduate earnings by major. The

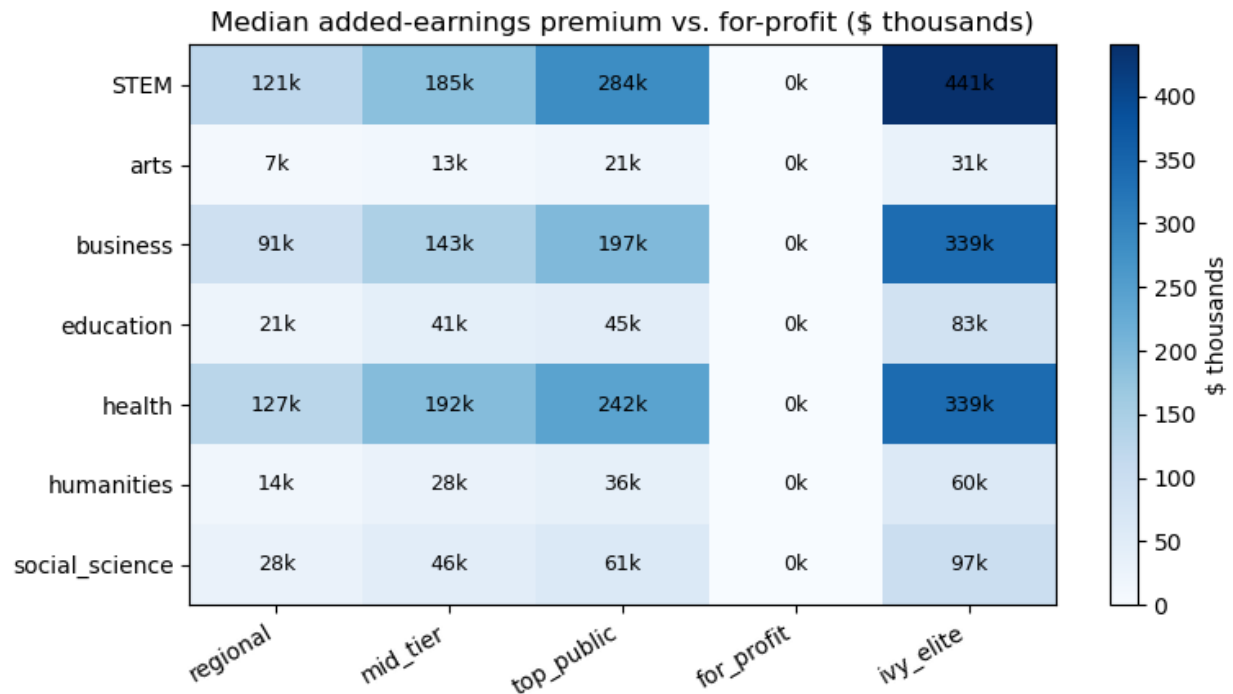
¹ https://www.kaggle.com/datasets/sergionefedov/college-major-roi?select=college_major_roi.csv

dataset is one row per person, containing the major, type of school, earnings, and post-graduate earnings. I compared the earnings by university tier and major. I used box plots to understand the spread. Below is figure 1.



I checked the balances of the majors by tier to search for any possible skew; these majors were very well balanced. The classes did not have any low major representation, so the major and university tier could be examined individually. I used OLS on major tier and both together to find the impact factor of each on earnings while adjusting for tier cost. The more expensive Ivy schools would have a higher model cost due to their actual cost to attend. So, to control for this cost, it was used as another predicting factor. The analysis for cost impact looked at the R-squared value before and after cost: before at .608 and after .609. Since the R value did not move at all, cost was determined to not be significantly involved.

I looked at the intersection of major and tier and a F test to explain the variance. I also visualized this with a heat map, shown below.



I used the F test to see if the university tier premium differs by major; there was a nominal dollar amount gained, but this gain was more prevalent in STEM, Business, and Health. I then looked further into this by adding in earnings as a log to see the multiplier effect. The Log revealed R-squared going from .61 to .82, suggesting this was a stronger correlation than before which was moderate at best.

Conclusion

It appears that certain majors have a greater influence on earning potential than school ranking. Major selection account for about 55% of variation on earnings for the 10-year time window and nearly 77% on the log scale. Institution was on 6% and 5% respectively. Another clear conclusion is that for-profit institutions do not have the payoff compared o the time one might spend in the program. This makes sense considering these programs are expensive and the higher cost hurts the payoff.

There is also the unaccounted human element. Someone coming from a more challenged economic background might be more inclined to try harder than someone of privilege or vice versa. The conclusion is limited, but revisiting this after more time of AI adoption would be interesting to see if behavior in selecting certain majors changes output.

References

https://www.kaggle.com/datasets/sergionefedov/college-major-roi?select=college_major_roi.csv